

Bottleneck-Relative Organization of Task-Relevant Phenomenal Content: The You Are Here Hypothesis

Abstract

Goal-directed action contains functionally distinguishable relations, while conscious task experience has structured focal content. This theoretical article addresses a general-psychology question about human subjectivity: whether the functional location at which goal-directed action is currently limited predicts what the focal experienced problem is about. The You Are Here hypothesis proposes that, within a substantially common action architecture, the identity of the currently limiting transition predicts task-relevant focal phenomenal content beyond generic demand and semantic reconstruction. The proposal remains empirically undemonstrated.

The minimal prediction is

$$I(B; \Phi_F | X, A) > 0.$$

The stronger dynamic prediction is

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j.$$

The dynamic prediction is distributional, not deterministic.

A future empirical strategy would manipulate or independently validate bottleneck location and test whether immediate report content tracks bottleneck identity beyond generic difficulty, attentional capture, metacognitive inference, explicit task-state diagnosis, demand characteristics, and semantic reconstruction. The proposal fails if those alternatives or existing process-specific accounts absorb the differentiation, or if focal content does not relocate when the bottleneck moves.

The contribution is a falsifiable cross-level correspondence linking functional action organization to lived task content without treating consciousness as moving through layers or impasse as creating consciousness.

Keywords: consciousness; human subjectivity; goal-directed action; bottleneck; phenomenology; metacognition; theoretical psychology

1. Introduction: The Missing Location Problem

Goal-directed action contains more causal structure than ordinarily appears in focal experience. Even a simple act may involve maintaining a goal, constructing candidates, comparing alternatives, specifying implementation, predicting consequences, monitoring feedback, and revising behavior. These relations can interact recurrently and hierarchically, yet when action becomes difficult the immediate experienced problem is often much narrower than this larger architecture. This mismatch raises a general-psychology

question about human subjectivity: how does the functional organization of goal-directed action relate to the organization of what is experienced as the problem?

At a given moment, the problem may concern missing possibilities, unresolved competition among alternatives, implementation, or whether an ongoing action is working. The larger system may involve all of these relations, but focal task experience need not represent them equally. Why should one portion of the action-generating process become the experienced problem rather than another?

The question is distinct from whether difficult or nonroutine processing recruits broader access or deliberation. Prior work links conscious access to global integration and demanding cognition, and goal obstruction to sustained focal attention to obstructed action plans (Baars, 2002; Bor & Seth, 2012; Dehaene & Naccache, 2001; Lewis & Todd, 2005). That general relationship is not the proposed novelty.

Nor does the question require conscious understanding of the processing architecture. Difficulty from a particular source can be phenomenally available without conscious identification of that source (Desender et al., 2021); subjective impasse reports need not align with behaviorally defined impasse stages, and impasse experience can be multidimensional and dynamic (Fedor et al., 2015; Ross & Arfini, 2024).

Suppose a common goal-directed architecture can be limited because viable candidates are absent, competition among available candidates is unresolved, or a selected candidate remains underspecified for implementation.

If the location of the limiting relation changes while the larger architecture remains substantially the same, does the functional content of the task-relevant phenomenal field change correspondingly?

In shorthand:

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j,$$

where B_i denotes the functional transition currently limiting progression and Φ_i denotes the task-relevant phenomenal problem associated with that unresolved relation. The implication is shorthand for a predicted shift in the distribution or referential organization of focal content when the dominant bottleneck changes; it is not a deterministic token-level law.

The proposal developed here is called **You Are Here** because it concerns location within a functional architecture. The metaphor should not be taken to imply that consciousness literally moves through processing layers, that consciousness is generated by impasse, or that all phenomenal content is determined by whichever relation is limiting.

The hypothesis instead concerns a possible cross-level correspondence:

functional location of unresolved progression

and

functional organization of the focal experienced problem .

Several weaker hypotheses could produce similar observations. Different bottlenecks could merely generate different degrees of generic difficulty:

$$B_i \rightarrow D \rightarrow \Phi_{difficulty}.$$

Alternatively, any sufficiently demanding bottleneck could recruit global access or cognitive control while leaving phenomenal content functionally undifferentiated. Process-specific metacognitive signals might also create differentiated feelings without implying a more general bottleneck-relative organizing principle. A further alternative is semantic reconstruction: the explicit local task state may itself tell a participant what currently requires work, allowing a condition-specific report to be constructed even if that relation was not phenomenally focal before the report probe.

You Are Here adds something only if bottleneck identity predicts what the task-relevant conscious problem is **about** beyond generic variables such as difficulty, effort, uncertainty, conflict, or time-on-task.

Let A denote a substantially common action architecture, B the functional identity of the currently limiting transition, Φ_F the task-relevant focal phenomenal field, and X a set of generic demand variables.

The minimal theoretical prediction is:

$$I(B; \Phi_F \mid X, A) > 0.$$

This paper does not report an experiment testing that prediction.

Its contribution is theoretical and integrative within general psychology. It brings together established work on functional differentiation in action generation, local impasse computation, conscious processing under unresolved conditions, phenomenal-field organization, and metacognitive difficulty in order to identify a narrower cross-level correspondence between functional action organization and lived task content that remains empirically open.

The argument proceeds in five steps.

First, action production is sufficiently functionally differentiated for bottleneck identity to be meaningful. Second, computational and problem-solving architectures show that different local failures can organize different local problem spaces. Third, established consciousness and control literatures already explain why unresolved processing may become globally or consciously relevant. Fourth, phenomenological and metacognitive work shows that the experienced task field can possess structured content without transparently revealing its computational cause. These premises together expose, but do not establish, the remaining correspondence.

The central question is therefore:

Move the bottleneck. Does the experienced problem move with it?

2. Functional Differentiation in Goal-Directed Action

The proposed variable B is meaningful only if goal-directed action can become limited at functionally distinguishable relations.

A large literature supports such differentiation without requiring a single proprietary processing pipeline.

2.1 Possibilities and candidate actions

The space of possible actions is generally larger than the subset represented as behaviorally relevant candidates. Decision making therefore cannot always be reduced to choosing among supplied alternatives; candidates may first have to be generated or constructed (Kaiser et al., 2013; Kalis et al., 2013).

Failure to generate a suitable candidate differs functionally from failure to choose among existing candidates. If no viable candidate is available, selection is premature; once several are represented, progression may still fail even though generation has succeeded.

The two problems can therefore be schematized as:

$$B_{generation} \neq B_{selection}.$$

This is a functional, not necessarily temporal, distinction: candidate construction and evaluation may interact. The point is that **absence of adequate candidates** and **unresolved competition among available candidates** are not identical computational conditions.

2.2 Selection and specification

A further distinction concerns the relation between choosing an action and specifying how that action will be implemented.

Affordance-competition approaches to action organization emphasize that multiple potential actions may be represented and compete within sensorimotor systems, with selection and specification emerging through distributed interactions rather than through a single central decision box (Cisek, 2007).

An action can therefore be sufficiently favored to count as selected while important parameters of its implementation remain unresolved.

A person may know what they intend to do while remaining uncertain how to perform it; specification may concern timing, trajectory, sequencing, or other context-sensitive parameters.

Thus:

$$B_{selection} \neq B_{specification}.$$

Again, this need not imply strict seriality. Specification can influence selection and unsuccessful implementation can reactivate earlier competition or candidate generation.

2.3 Monitoring, feedback, and revision

Goal-directed action also depends on monitoring consequences, discrepancies, environmental change, and whether ongoing behavior continues to serve the governing goal.

A failure to determine whether an action is succeeding presents a different functional problem from failure to generate, select, or specify an action in the first place.

Schematically:

$$B_{monitoring} \neq B_{generation}, B_{selection}, B_{specification}.$$

The present paper does not require these four labels to exhaust action organization. They are illustrative functional relations within a much richer architecture.

2.4 Functional identity without a rigid pipeline

These distinctions should not be converted into a rigid serial model.

Goal-directed behavior can be hierarchically organized, and the functional relations considered here need not operate as a single flat sequence (Botvinick, 2008). Selection at one level may constrain candidate generation at another. Attempted implementation can reveal that a previously selected action is infeasible. Monitoring can reopen selection. A new environmental event can alter the governing goal before lower-level execution completes.

The relevant structure is therefore not:

$$\text{generation} \rightarrow \text{selection} \rightarrow \text{specification} \rightarrow \text{monitoring}$$

as an obligatory sequence.

It is instead a set of functionally distinguishable relations that may become locally limiting within an organized goal-directed system.

This permits a more general definition:

$$B = \text{functional identity of the relation currently limiting progression}.$$

Under this definition, a bottleneck is not necessarily a literal processing stage or anatomical location. It is a relation whose unresolved state prevents or materially constrains continued progression toward the current goal.

This distinction provides the first premise required by You Are Here:

“Unable to proceed” does not uniquely identify why progression is limited.

Different failures can create different local functional problems even within the same larger action architecture.

The next question is whether systems can organize processing around the identity of such local failures.

3. Local Failure and Impasse-Relative Computation

A useful precedent comes from computational architectures that represent different impasses and organize subsequent computation around the particular obstruction rather than a single generic failure state.

3.1 Impasse in Soar

Soar provides an especially clear example (Laird, 2012).

In Soar, different failures produce different impasses, including cases where no suitable operator can be selected, multiple operators remain unresolved, or a selected operator cannot be applied.

When an impasse occurs, Soar creates a substate in which processing is directed toward resolving that impasse. When the impasse is resolved and progression becomes possible again, the temporary substate is removed.

The relevant structural relation is:

specific impasse → specific temporary computational problem-space .

This demonstrates the computational coherence of local failure organizing a local problem: a system need not respond to every obstruction with an undifferentiated state, because the relation preventing progression can organize what must be solved next.

3.2 What the analogy does not establish

The boundary here is essential:

Soar substate ≠ human phenomenal consciousness .

Soar does not provide evidence that human conscious experience is organized by bottleneck location, nor does the argument require humans to implement Soar's mechanisms. Its role is limited to computational coherence.

The mapping

$$B_i \rightarrow P_i$$

where P_i is a temporary problem-space specifically organized around resolving B_i , is a viable form of system organization.

What remains unknown is whether an analogous relation exists at the level of task-relevant phenomenal organization:

$$B_i \overset{?}{\rightarrow} \Phi_i.$$

3.3 Human impasse and problem restructuring

Human problem solving also provides a broader context in which local obstruction reorganizes processing.

An impasse is not merely a stronger version of normal progress. In insight problem solving, overcoming impasse can require a change in problem representation, including relaxation of constraints or decomposition of perceptual chunks (Knoblich et al., 1999).

This provides another precedent for treating unresolved action as locally organized rather than globally homogeneous.

Yet the same evidentiary firewall applies.

Evidence that a problem solver changes strategy at an impasse does not demonstrate that the content of phenomenal experience tracks the functional identity of that impasse.

The computation may reorganize without the proposed phenomenal correspondence.

Section 3 therefore establishes only this:

specific functional obstruction can coherently organize a specific local computational problem.

Whether the phenomenal field exhibits a corresponding organization remains an independent empirical question.

4. Conscious Processing Under Unresolved Conditions

The claim that unresolved or difficult processing can become consciously important is not new.

Indeed, conceding this prior art is necessary to isolate the narrower contribution of You Are Here.

4.1 Global availability and nonroutine processing

Global Workspace and Global Neuronal Workspace approaches associate conscious access with broader availability of information across otherwise specialized processing systems (Baars, 2002; Dehaene & Naccache, 2001).

Within these traditions, globally accessed information becomes available for flexible report, deliberation, control, memory, and cross-functional coordination, contrasting with processing that can proceed without broad conscious access (Dehaene & Naccache, 2001).

Related theoretical work connects consciousness to working-memory integration and structured processing, while work on goal obstruction describes prolonged episodes in which attention can remain focused on action plans and their obstruction (Bor & Seth, 2012; Lewis & Todd, 2005).

None of this is the proposed novelty.

You Are Here does not claim:

breakdown $\rightarrow$ consciousness.

It does not claim that conscious processing first appears only when automatic processing fails.

It does not claim that every bottleneck becomes conscious.

And it does not claim that consciousness exists in order to resolve bottlenecks.

4.2 Cognitive control and conflict

Conflict-monitoring and cognitive-control work likewise supports the proposition that processing conflict can signal a demand for additional control (Botvinick et al., 2001).

These mechanisms can explain why unresolved processing receives greater priority without necessarily predicting the functional identity of phenomenal content.

A generic control account could accommodate:

$B_{generation} \rightarrow$ increased control

and

$B_{selection} \rightarrow$ increased control

without predicting any corresponding distinction between:

$\Phi_{generation}$

and

$\Phi_{selection}$.

Likewise, two unresolved conditions becoming globally available does not imply different task-relevant phenomenal organization according to obstruction location.

4.3 Consciousness is not required for every control effect

The manuscript must also avoid the opposite overstatement.

A considerable amount of processing can occur without consciousness, and processing can influence subjective difficulty even when its causal source is not consciously identified (Dehaene & Naccache, 2001; Desender et al., 2021).

This is especially important because the proposed hypothesis is not a boundary between conscious and unconscious processing.

The question is not:

At what stage does consciousness enter?

It is:

Conditional on there being task-relevant phenomenal content, does its functional organization correspond systematically to the location of unresolved progression?

This difference separates You Are Here from theories attempting to specify a universal functional role or necessary condition for consciousness.

4.4 What prior work leaves open

Existing literature therefore explains why:

- some tasks requiring novel combinations or durable maintenance depend on conscious access (Dehaene & Naccache, 2001);
- processing conflict can recruit additional control (Botvinick et al., 2001);
- conscious access can support integration across otherwise separate functions (Baars, 2002);
- persistent goal obstruction can sustain focal attention to action plans and their obstruction (Lewis & Todd, 2005);
- subjective difficulty can integrate multiple cues from ongoing performance (Desender et al., 2017).

But these propositions remain compatible with several different phenomenal outcomes.

All bottlenecks might converge on a common phenomenal state:

$$\Phi_{difficulty}.$$

Alternatively, each bottleneck might preferentially organize focal content around its own unresolved relation:

$$B_i \rightarrow \Phi_i.$$

Global-access and control theories do not automatically decide between those alternatives.

This is the boundary at which the phenomenal organization of the problem becomes important.

5. Organization of the Phenomenal Field

The second major premise is that conscious task experience is not merely scalar. Beyond degrees of uncertainty, effort, frustration, or confidence, experience has content: particular possibilities, objects, actions, discrepancies, or questions can become focal.

5.1 Theme, field, and margin

Phenomenological work associated with Gurwitsch distinguishes a focal theme from a thematic field and a broader margin (Gurwitsch, 1964; Yoshimi & Vinson, 2015).

The relevance of this framework is structural.

At any moment, experience can be organized around something focal without everything else disappearing from consciousness. The focal theme derives significance from its relation to a wider organized field.

This offers a useful vocabulary for the present problem because You Are Here does not propose that bottleneck location determines all phenomenal content.

It concerns:

$$\Phi_F = \text{task-relevant focal phenomenal field.}$$

Background bodily sensations, mood, environmental awareness, autobiographical thought, and unrelated conscious content can remain present.

The hypothesis concerns which task-relevant relation becomes focal.

5.2 Problem-solving phenomenology

Phenomenological analysis of problem solving further supports the idea that experienced problem structure can reorganize dynamically (Yoshimi, 2017).

Problem experience need not remain invariant through resolution: salience, unresolved relations, and relevant possibilities can change as solving progresses. This is compatible with the model's dynamic structure but does not establish the correspondence, because phenomenal fields can change for many other reasons.

5.3 Fields of relevant affordances

Skilled Intentionality likewise treats action relevance as selective rather than an undifferentiated inventory: certain affordances become relevant or solicit action in the current situation (Rietveld et al., 2018). This is a structural precedent for selective action relevance, not evidence for the You Are Here phenomenal-content correspondence.

5.4 Metacognitive feelings

Research on metacognition shows that subjective processing difficulty can be constructed from multiple cues arising during ongoing performance (Desender et al., 2017).

These findings create both support and competition for You Are Here.

They support the possibility that internal processing conditions systematically influence conscious experience.

But they also create a strong generic-demand alternative.

Perhaps different bottlenecks produce nothing more than different magnitudes or mixtures of:

difficulty , uncertainty , effort , conflict .

If so, a bottleneck-relative theory of content is unnecessary.

Recent metacognitive theories provide a substantially stronger comparator than scalar difficulty alone. Fernández Velasco and Loev (2025) characterize metacognitive feelings as conscious affective experiences concerning mental processes and capacities. In their predictive-processing account, metacognitive feelings arise from process-related prediction dynamics, can phenomenally appear directed either at a cognitive process or at the content toward which that process is directed, form dynamic descriptive and directive models, and guide mental action. Proust (2024) likewise treats metacognitive feelings as cognitive-affordance sensings that can regulate cognitive action at distinct control junctures, including selection or engagement, revision or correction during activity, and post-evaluation of outcomes.

These accounts provide substantial prior art for any broad claim that different cognitive processes can generate different conscious metacognitive feelings, that such feelings can be content-directed and dynamically updated, or that they can guide cognitive control. You Are Here therefore cannot claim those propositions as its distinctive contribution.

The proposed correspondence is narrower. It asks whether an **independently identified dominant relation limiting progression within a substantially common action architecture** predicts the referential organization of task-relevant focal phenomenal content beyond generic demand and beyond what explicit task-state knowledge could generate at report time, including a distributional relocation of focal content when the dominant bottleneck moves. Neither comparator, as formulated in the sources cited here, states that full empirical localization criterion. Whether their frameworks can nevertheless derive the complete pattern is treated as a live subsumption test rather than assumed away.

The proposed correspondence therefore cannot be supported merely by showing that people feel more difficulty under one condition than another, or merely by showing process-specific metacognitive feelings.

5.5 Difficulty without transparent causal access

Desender and colleagues provide an especially important constraint (Desender et al., 2021).

Their work indicates that action-selection difficulty induced by subliminal response conflict can be consciously experienced even when participants are unaware of the prime producing that conflict.

This motivates a three-way distinction:

B = functional bottleneck ,

Φ_F = task-relevant phenomenal content ,

and

M = explicit metacognitive diagnosis .

The hypothesis does not require:

$$B = \Phi_F = M .$$

A participant might experience uncertainty organized around competing alternatives without consciously knowing which lower-level processing event produced that uncertainty.

Conversely, an explicit verbal explanation of being “stuck at selection” would not by itself prove that selection-related content structured the immediate phenomenal field.

5.6 Subjective impasse is not objective diagnosis

The Five-Square impasse literature provides a related warning (Fedor et al., 2015).

Subjective reports of feeling stuck did not reliably align in time with behaviorally defined stages of impasse.

Ross and Arfini further characterize impasse phenomenology as multidimensional and dynamic (Ross & Arfini, 2024).

These findings rule out a simplistic prediction:

$$B_i \rightarrow \text{explicit awareness that “I am objectively stuck at } B_i \text{.”}$$

That is not You Are Here.

The relevant prediction concerns the referential organization of the task problem, not the accuracy of architectural introspection.

A participant can fail to diagnose why progression is blocked while still having experience organized around particular alternatives, implementation details, feedback relations, or another task object.

5.7 Structured content without architectural transparency

Sections 2–5 therefore establish two different forms of organization.

The action architecture contains functionally distinguishable relations.

The phenomenal field can contain functionally distinguishable task-relevant content.

But the latter need not transparently represent the former.

This distinction is crucial.

If phenomenal experience simply mirrored the processing architecture, the proposed hypothesis would be trivial. If phenomenal experience contained only scalar demand states, the hypothesis would likely be false.

The unresolved possibility lies between those extremes.

The phenomenal field may selectively expose a problem relation without providing a complete or explicit model of why that relation has become problematic.

That possibility leads to the central correspondence.

6. The Untested Correspondence

The preceding literatures establish two forms of organization that are usually studied separately.

On the functional side, goal-directed action can be decomposed into distinguishable relations involved in generating, selecting, specifying, monitoring, and revising action. These relations need not form a rigid serial sequence, and their boundaries need not correspond to discrete neural modules. It is sufficient that progression can be limited for functionally different reasons.

A system that lacks viable candidates is not in the same computational predicament as one that has several candidates but cannot resolve among them, and neither is identical to a system that has selected an action but cannot determine how to implement it.

Research on impasse and cognitive architecture further shows that local failures can organize local computation. In architectures such as Soar, different impasses can induce temporary problem spaces directed toward resolving the particular relation that prevents progression.

This establishes the computational coherence of an impasse-relative organization of processing.

It does not establish an analogous organization of human phenomenal experience.

A computational substate is not a phenomenal field, and no such identification is required by the present argument.

On the phenomenal side, several traditions likewise reject the idea that conscious task experience is an undifferentiated scalar.

Phenomenological accounts distinguish a focal theme from a wider field and margin; work on skilled intentionality emphasizes structured fields of relevant affordances; research on metacognitive feelings and problem-solving impasse shows that experienced difficulty can possess multiple dimensions and can change dynamically during an unresolved task.

Importantly, phenomenally available difficulty need not reveal its true computational cause.

A person may experience that something about ongoing action is difficult without possessing an accurate metacognitive diagnosis of the process responsible for that difficulty.

These observations motivate the distinction:

B = the functionally limiting relation ,

Φ_F = the task-relevant focal phenomenal field ,

and

M = an explicit metacognitive diagnosis of the difficulty .

The present proposal does not assume:

$$B = \Phi_F = M.$$

In particular, it does not require a person to know which component of an action architecture is objectively limiting.

Nor does it require the phenomenal field to reproduce the architecture itself.

The relevant question is narrower:

Does the functional identity of the unresolved relation predict the functional identity of what the focal task problem is about?

Nothing established so far licenses that inference.

Existing theories can explain why novelty, ambiguity, conflict, effort, or obstruction increase controlled processing or conscious relevance.

Existing phenomenological and metacognitive accounts can explain why task experience has structure and why unresolved situations feel difficult in different ways.

Existing action theories can distinguish candidate generation, selection, specification, monitoring, and related functions.

These results do not by themselves establish that relocating the functionally limiting relation within a substantially common action architecture correspondingly reorganizes the content of the focal phenomenal problem.

Let A denote the relevant goal-directed action architecture, B the functional identity of the currently limiting transition, Φ_F the task-relevant focal phenomenal field, and X a set of generic demand variables including difficulty, uncertainty, effort, conflict, and time-related demand.

The unresolved correspondence can then be stated as:

$$I(B; \Phi_F \mid X, A) > 0.$$

This expression does not assert that the present manuscript estimates conditional mutual information. It is a compact dependence and predictive criterion, not by itself a causal estimand. Conditional mutual information is symmetric and therefore does not establish the direction of influence between B and Φ_F . Any stronger causal interpretation would require additional identification from the experimental manipulation and design.

The expression formalizes the minimal empirical distinction at issue. If bottleneck identity provides no information about phenomenal-content identity once generic demand and the larger architecture are accounted for, then the proposed correspondence adds no explanatory information. Conversely, satisfying this statistical criterion alone would not establish phenomenal organization, because condition-specific reports could still be reconstructed from explicit task-state semantics. That distinct identification problem is addressed in Sections 8 and 9.

The critical contrast is therefore not between easy and difficult processing, or between automatic and consciously controlled processing.

It is between two possibilities.

Under a generic-demand account, functionally different bottlenecks may differ in intensity while converging phenomenally on a common state such as effort, uncertainty, confusion, or undifferentiated stuckness.

In that case, once those generic variables are known, bottleneck identity should provide little or no further information about focal phenomenal content.

A different possibility is that the identity of the unresolved relation systematically predicts the organization of the experienced problem itself.

Under that possibility, difficulty associated with insufficient candidate generation should not be phenomenally equivalent, in referential structure, to equally demanding difficulty associated with unresolved selection or action specification.

The distinction would concern what the problem is about, not merely how strongly it is felt.

The remainder of this paper refers to this proposed correspondence as **You Are Here**.

The name denotes a location claim within an action architecture, not a claim about the quantity, origin, or function of consciousness.

The hypothesis is introduced in the next section as a testable organizing principle.

It has not yet been empirically demonstrated.

7. You Are Here: Bottleneck-Relative Phenomenal Organization

The You Are Here hypothesis proposes that, within a substantially common goal-directed action architecture, the functional identity of the currently limiting relation predicts the organization of task-relevant phenomenal content.

In its minimal form:

$$I(B; \Phi_F \mid X, A) > 0.$$

The prediction is not that bottlenecks produce consciousness, that unresolved processing necessarily becomes conscious, or that greater difficulty produces more consciousness.

It is an identity correspondence. The conditional-information expression above states a residual predictive dependence; it does not, by itself, establish causal direction.

Holding the larger architecture and generic demand as constant as practicable, knowing where progression is functionally limited should provide information about what the focal experienced problem is about.

The stronger prediction is dynamic:

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j.$$

If the dominant limiting relation changes during ongoing engagement with a common task architecture, the distribution or referential organization of the task-relevant phenomenal field should shift correspondingly. The notation is therefore probabilistic and population/distributional in interpretation: it does not require a one-to-one phenomenal transition on every trial or for every individual.

This is the central empirical commitment of the model.

A static association between a particular task and a particular kind of experience would be insufficient because such an association could reflect stable task features rather than bottleneck-relative organization.

Consider a simplified action architecture in which progression can be limited at several functionally distinguishable relations.

If candidate generation is limiting, the unresolved problem concerns the availability or construction of viable possibilities.

If several viable candidates are available but progression is limited by unresolved competition among them, the problem concerns their comparative relation.

If a candidate has been selected but progression is limited by implementation, the unresolved problem concerns specification of how the action is to be carried out.

If execution is under way but progression is limited by uncertainty about its effects, the unresolved problem concerns feedback, discrepancy, or whether the current action is working.

Illustratively:

$$B_{generation} \rightarrow \Phi_{available\ possibilities},$$

$$B_{selection} \rightarrow \Phi_{relations\ among\ alternatives},$$

$$B_{specification} \rightarrow \Phi_{implementation},$$

and

$$B_{monitoring} \rightarrow \Phi_{feedback/discrepancy}.$$

In ordinary language, these might resemble differences among:

“What can I do?”

“Which one?”

“How do I do it?”

“Is this working?”

These formulations are explanatory illustrations only.

They are not proposed as universal introspective sentences, fixed phenomenal categories, or empirical findings.

The hypothesis also does not require explicit access to the architecture generating the experience.

A person need not know that candidate generation, selection, or specification is the objectively limiting relation.

Indeed, the distinction among B , Φ_F , and explicit metacognitive diagnosis M predicts that causal diagnosis may be inaccurate while task-relevant phenomenal content remains systematically differentiated.

The experienced problem may concern alternatives without the person possessing the concept “selection bottleneck,” just as consciously experienced difficulty can be influenced by processing whose causal source is not consciously accessible.

This distinction matters because a theory equating phenomenal correspondence with accurate introspective diagnosis would be unnecessarily strong and inconsistent with existing evidence on metacognitive opacity and subjective impasse.

You Are Here instead concerns the referential organization of the focal problem: what possibilities, relations, implementation details, discrepancies, or other task objects currently structure experience.

Nor does the proposal require action generation to proceed through a fixed series of discrete stages.

B denotes the functional identity of a limiting relation, not the ordinal location of a box in a pipeline.

Real action systems may be recurrent, hierarchical, partially parallel, and dynamically coupled.

Candidate generation may alter selection; attempted specification may reveal the need for new candidates; monitoring may reopen earlier alternatives.

The hypothesis requires only that limiting relations can sometimes be distinguished sufficiently to support meaningful intervention or identification.

The same restriction applies to multiplicity.

The present formulation does not assert that exactly one bottleneck exists at every moment.

Multiple constraints may coexist.

The strongest initial test concerns conditions in which one relation can be made dominant enough for its functional identity to be experimentally manipulated and independently verified.

More distributed cases would require a richer formulation and are treated here as a scope issue rather than as evidence for or against the simplest model.

Most importantly, You Are Here is not a theory of phenomenal intensity.

A generation bottleneck need not feel harder than a selection bottleneck, nor should stronger limitation necessarily produce a larger quantity of conscious experience.

Two conditions could be closely matched in rated difficulty and still differ in the referential organization of the phenomenal field.

The central claim can therefore be stated without metaphor:

The functional identity of a currently limiting relation within goal-directed action may systematically predict the

The word *may* is essential.

The correspondence remains a theoretical prediction.

Its value depends on whether it yields observations that competing explanations do not already predict and whether those observations survive attempts to separate content identity from generic demand.

8. Alternative Explanations and Discriminating Predictions

The You Are Here hypothesis is useful only if it predicts something beyond established accounts of difficulty, control, conscious access, and process-specific metacognition.

Several alternatives can reproduce parts of the pattern that might superficially appear to support the proposal.

The relevant question is therefore not whether bottleneck manipulations alter experience, but what kind of alteration occurs and whether bottleneck identity explains anything after stronger alternatives are considered.

8.1 Generic-demand account

The simplest alternative is that functionally different bottlenecks differ only in generic demand.

Let D denote a latent demand state expressed through variables such as subjective difficulty, uncertainty, effort, conflict, confusion, or prolonged response time.

Subjective difficulty can itself integrate multiple cues from ongoing performance, making this alternative more than a simple nuisance explanation (Desender et al., 2017).

A generic-demand model can be represented schematically as:

$$B_i \rightarrow D \rightarrow \Phi_{\text{difficulty}}.$$

Under this account, different bottlenecks can produce different behavioral consequences and different degrees of consciously experienced difficulty without producing different kinds of focal phenomenal content.

A participant facing inadequate candidate generation and a participant facing unresolved selection may both simply experience greater uncertainty or effort.

This alternative predicts that once generic demand is accounted for:

$$I(B; \Phi_F | X, A) \approx 0.$$

Such a result would directly undermine the You Are Here hypothesis.

Reports such as “this is difficult,” “I am confused,” “I am stuck,” or “I am unsure” therefore cannot, by themselves, count as evidence for bottleneck-relative phenomenal organization.

They establish difficulty, not functional content.

The critical positive prediction is instead referential differentiation: reports should differ systematically in what the unresolved problem concerns even when generic demand is matched or statistically controlled.

8.2 Global-access and cognitive-control accounts

A second class of alternatives explains why disrupted or unresolved processing becomes behaviorally and consciously important.

Workspace accounts provide mechanisms for broad information availability, while conflict-monitoring accounts provide a mechanism for recruiting control under processing conflict (Baars, 2002; Botvinick et al., 2001; Dehaene & Naccache, 2001).

You Are Here does not compete with those accounts over the existence of such recruitment.

It asks a different question.

A global-access account could predict:

$$B_i \rightarrow \text{increased access/control}$$

for many different B_i without predicting:

$$B_i \rightarrow \Phi_i.$$

If generation, selection, and specification bottlenecks all increase access or control but produce only a common phenomenal state of greater difficulty, salience, or attentional capture, the result would support a global-recruitment account without supporting You Are Here.

Evidence favoring the present hypothesis would therefore require content differentiation beyond the degree of access, salience, attentional capture, effort, or control recruitment.

8.3 Process-specific metacognitive accounts

A stronger alternative is that different cognitive processes generate different metacognitive feelings. This is not merely a hypothetical competitor. Fernández Velasco and Loev (2025) propose a predictive-processing account in which metacognitive feelings are conscious affective experiences concerning particular mental processes and capacities. Their account allows a feeling to appear phenomenally directed either at the process itself or at the content toward which that process is directed, treats metacognitive feelings as dynamic models that are both descriptive and directive, and assigns them a role in shaping mental-action policies. Proust (2024) similarly treats metacognitive feelings as cognitive-affordance sensings involved in selecting or engaging cognitive actions, revising or correcting ongoing activity, and evaluating outcomes.

These accounts overlap substantially with the territory surrounding You Are Here. They establish that process-specific, content-directed, dynamically updated conscious feelings can participate in cognitive control. A finding that generation, selection, specification, or monitoring conditions merely produced different metacognitive feelings would therefore not by itself distinguish the present proposal from this prior art.

The residual You Are Here claim is more specific and relational. It predicts that, **within a substantially common action architecture, an independently identified dominant relation limiting progression predicts the referential organization of task-relevant focal phenomenal content beyond generic demand and beyond task-state/report reconstruction, with a corresponding distributional relocation when the dominant bottleneck moves.**

This formulation does not establish novelty against the comparator accounts. Rather, it identifies the additional empirical pattern that the present manuscript requires. If the Fernández Velasco and Loev or Proust frameworks, or another established metacognitive account, can derive the full pattern under the same identification conditions, including the dominant-bottleneck relation and relocation result, then the present proposal is absorbed to that extent and should be reduced or eliminated accordingly.

8.4 Task-state semantic reconstruction

A distinct alternative can generate the predicted report content even when pre-prompt phenomenal focality does not differ by bottleneck. The objective local task state may make a particular relation explicitly relevant: a generation-limited condition makes the availability of possibilities objectively problematic, a selection-limited condition makes relations among alternatives objectively problematic, and a specification-limited condition makes implementation information objectively problematic. A competent participant may therefore infer what currently requires work and produce a condition-specific report from that explicit structure.

The competing route is:

explicit local task state / problem structure → inference about what currently requires work → condition-specific
without requiring:

$$B \rightarrow \Phi_F \rightarrow \text{report}.$$

This alternative is not another scalar component of generic demand. It can survive matching or statistical control for difficulty, uncertainty, effort, conflict, and response latency, and it can persist within a common architecture. It can also mimic the dynamic prediction because the explicit local task state may change when the dominant bottleneck changes. The same logic covers ordinary demand-characteristic effects if participants infer what kind of response the experimental condition invites.

Simply adding every local task-state variable to X would not solve the problem and could condition away part of the substantive relation under study. The distinction instead requires design-based identification. A positive result supports bottleneck-relative **reportable task representation** if the same report structure can be generated from explicit task-state diagnosis or from control conditions preserving the relevant task-state semantics. Evidence more specifically consistent with bottleneck-relative phenomenal organization requires that immediate report structure track independently validated bottleneck dominance in ways not fully reproduced by those semantic-reconstruction routes.

If task-state/report reconstruction or demand characteristics fully explain the positive pattern, the phenomenal interpretation is not identified and the strong You Are Here claim is not supported.

8.5 Retrospective reconstruction

Another alternative is that participants construct a coherent explanation only after the unresolved episode has passed.

Suppose a task condition objectively limits candidate generation.

After completion, a participant may infer that the problem must have been “not knowing what to do,” even if no such differentiated content organized experience during the episode itself.

Post-hoc reports could therefore generate an apparent B -to- Φ mapping through narrative reconstruction.

For this reason, the strongest form of You Are Here predicts differentiation in reports sampled during or immediately adjacent to unresolved processing.

A correspondence that appears only after extended reflection would not establish online bottleneck-relative phenomenal organization.

If the mapping disappears under temporally proximal measurement and emerges only retrospectively, the model’s strong phenomenal claim should be rejected or substantially weakened.

8.6 Architectural confounding

A further alternative is that apparent phenomenal differentiation reflects differences among tasks rather than differences in bottleneck location.

Comparing an unconstrained brainstorming task with a forced-choice task and a motor-execution task would almost guarantee differences in reported experience.

Such a comparison would not establish You Are Here because the manipulation changes much more than the limiting relation.

The relevant test requires the larger action architecture to remain substantially common while the experimentally limiting relation is shifted within it.

The ideal contrast is therefore:

$$A \approx \text{constant}, B_i \rightarrow B_j.$$

The closer a design approaches “different tasks produce different experiences,” the less diagnostic its results become.

8.7 Dynamic discrimination

The strongest prediction distinguishes You Are Here from explanations based on stable task identity or retrospective categorization.

If, during a continuing task, the limiting relation moves from B_i to B_j , the model predicts:

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j.$$

A participant might first lack viable possibilities, later generate several but be unable to choose among them, and subsequently select one while remaining uncertain how to implement it.

Under the hypothesis, those transitions should produce a corresponding shift in the distribution or referential organization of focal content rather than leave the phenomenal field fixed on the initial difficulty state. The prediction is not that every token transition must yield an identical one-to-one report change.

Failure of distributional relocation in a clean design is a substantive negative result.

If bottleneck identity changes while the functional organization of the focal phenomenal field remains stable at the relevant aggregate or within-person level, the strongest form of the model is contradicted.

8.8 Summary of discriminating predictions

Account	Primary predicted consequence of bottlenecking	Result required specifically by You Are Here
Generic demand	Greater difficulty, uncertainty, effort, conflict, or confusion	Referential content differs after generic demand is accounted for
Global access / control	Greater access, salience, or controlled processing	Content identity varies with bottleneck identity rather than merely access intensity
Process-specific metacognition	Conscious, process-specific, content-directed feelings that guide cognitive control	The full independently validated dominant-bottleneck relation survives established metacognitive predictions and the other identification controls
Task-state semantic reconstruction	Explicit local problem structure determines what the participant can infer and report	Immediate report content tracks validated bottleneck dominance beyond explicit task-state diagnosis and semantic-yoked controls

Account	Primary predicted consequence of bottlenecking	Result required specifically by You Are Here
Retrospective reconstruction	Differentiated post-hoc explanations	Differentiation is present during or immediately adjacent to unresolved processing
Task-identity confound	Different tasks produce different reports	Bottleneck location changes within a substantially common architecture
You Are Here	Bottleneck-relative organization of focal phenomenal content	$I(B; \Phi_F X, A) > 0$, plus identification beyond task-state reconstruction and distributional relocation where testable

The model therefore volunteers six strong failure conditions.

First, if bottleneck manipulations alter only generic difficulty or related demand variables, the model fails.

Second, if condition-specific reports can be fully explained by explicit task-state knowledge or report reconstruction, the phenomenal interpretation is not identified and the strong model is unsupported.

Third, if differentiated phenomenal content appears only through retrospective explanation, the strongest online interpretation fails.

Fourth, if the relevant functional relations cannot be independently distinguished well enough to identify B , the model is empirically underdetermined.

Fifth, if the strongest metacognitive or other established theories fully predict the observed differentiated pattern under the same identification conditions, the present proposal adds nothing beyond those accounts.

Sixth, and most directly, if bottleneck location changes during a continuing task while the distribution or referential organization of focal phenomenal content does not relocate correspondingly, the strong dynamic prediction is contradicted.

These failure conditions are not auxiliary qualifications.

They define what would make the proposed correspondence scientifically informative.

The next section therefore turns from conceptual discrimination to a proposed empirical strategy capable, in principle, of producing either the predicted correspondence or one of these negative outcomes.

9. Empirical Strategy

The You Are Here hypothesis is intended to be empirically vulnerable.

Its strongest test requires more than showing that different tasks produce different subjective reports or that greater difficulty produces greater conscious effort.

The relevant experiment must manipulate the functional location of a limiting relation while preserving as much of the larger action architecture as possible, and it must measure phenomenal-content structure independently of the labels used to define the manipulation.

No such experiment is reported here.

The following design is proposed as a future test.

9.1 Moving the limiting relation

A useful paradigm would employ a common goal-directed task architecture in which progression depends on at least three distinguishable functional relations:

candidate generation ,
selection among candidates ,
and
action specification .

The experimental manipulation would selectively make one of these relations more limiting while leaving the larger task, goal, response structure, and other surface properties as similar as practicable.

In a generation-limited condition, progression should be constrained primarily by the availability or construction of viable candidates.

In a selection-limited condition, viable candidates should be available but progression should be constrained primarily by unresolved competition among them.

In a specification-limited condition, a candidate should be sufficiently selected while progression is constrained primarily by how the intended action is to be implemented.

These descriptions define functional targets, not rigid processing stages.

A successful paradigm would require independent evidence that the manipulation actually changes the dominant limiting relation rather than merely changing task difficulty.

The critical intervention is therefore:

$$A \approx \text{constant} ,$$

while:

$$B_{\text{generation}} \rightarrow B_{\text{selection}} \rightarrow B_{\text{specification}}$$

is experimentally varied.

The closer the design comes to changing only the functionally limiting relation, the more informative the comparison becomes.

9.2 Identifying the bottleneck independently

Bottleneck identity cannot be inferred from the phenomenal report used to test the hypothesis.

Doing so would make the proposed correspondence circular.

The experimental manipulation must therefore establish B independently through task structure and appropriate behavioral signatures.

The exact signatures will depend on the eventual paradigm, but the inferential order must remain:

experimental manipulation $\rightarrow B$

independently of:

phenomenal report $\rightarrow \Phi_F$.

If a manipulation intended to isolate candidate generation cannot be distinguished behaviorally from one that increases general uncertainty or later selection difficulty, the condition cannot provide a clean test of the model.

Architectural identifiability is therefore a prerequisite rather than an assumption.

A preregistered factor-contraction rule should govern cases in which the intended bottleneck classes are not independently separable. If two candidate relations cannot be discriminated by task structure and the prespecified behavioral validation criteria before report analysis, they must be collapsed to the coarsest functional distinction that is independently identifiable for that paradigm, or the ambiguous trials must be excluded from the clean test. A finer label may not be retained merely because the theory names it. The corresponding hypothesis test must then use the contracted factor. This rule prevents an apparent negative result from being rescued post hoc by preserving distinctions that the experiment did not establish.

9.2.1 Worked hypotheticalal common-architecture paradigm

One concrete implementation can illustrate how these requirements could be realized without implying that the manipulations have already been validated in human participants. Consider a computerized route-planning task in which every trial has the same goal: move a token from a start state to a target state under a fixed family of movement rules. The interface, response method, route workspace, and final execution controls remain constant. What changes is the information structure intended to make candidate generation, selection among viable routes, or specification of the chosen route the dominant constraint. The example below is a proposed design, not participant data, and the manipulation counts as a test of B only if the prespecified task and behavioral criteria are actually satisfied.

Condition / manipulation	Independent B-validation target	Common measurements	Predicted / diagnostic outcome
Generation-limited: the map and movement rules remain visible, but no candidate route is supplied and trial constraints make viable routes sparse enough that the participant must construct one.	Before report analysis: delayed first viable candidate and/or low valid-candidate yield, followed by rapid downstream choice and specification when the system supplies a valid candidate on matched rescue	Difficulty, uncertainty, effort, conflict, response latency; the same unpredictable neutral probe; condition-blind coding of R_F ; explicit task-state diagnosis on separate trials/blocks.	If B is generation-limited, report content should shift toward missing/constructing possibilities. Generic difficulty alone is insufficient; no support if the condition cannot be separated from selection or if

Condition / manipulation	Independent B-validation target	Common measurements	Predicted / diagnostic outcome
	trials. Failure of the rescue pattern means generation is not cleanly established.		semantic controls reproduce the same report pattern without generation dominance.
Selection-limited: multiple viable routes are present and comparably attractive, so progression depends on resolving among already available candidates.	Before report analysis: at least two candidates independently satisfy the viability criterion, choice is prolonged or alternatives are repeatedly revisited, and externally designating one viable route releases the delay while specification remains near the non-limited baseline.	The same demand ratings, latency logging, neutral probe, blind R_F coding, and separate explicit task-state diagnosis.	If B is selection-limited, report content should shift toward relations or competition among alternatives. No support if candidate availability was not established, if designation does not release the delay, or if explicit task semantics alone generate the same pattern.
Specification-limited: one route is clearly preferred or is fixed in advance, but trial-specific control mappings or implementation parameters needed to execute that route remain unresolved.	Before report analysis: route choice is stable with little meaningful candidate switching, while choice-to-executable-plan latency and/or implementation errors are elevated; revealing the missing mapping releases the delay without changing the selected route.	The same demand ratings, neutral probe, blind R_F coding, and separate task-state diagnosis; matched rescue trials verify that the mapping rather than route choice released progression.	If B is specification-limited, report content should shift toward how to implement the chosen route. No support if selection remains unstable, if mapping information does not release the delay, or if the same report structure occurs in yoked trials without specification dominance.

The same neutral report probe would be used across conditions: “What were you trying to figure out immediately before this prompt?” Reports would be pooled without condition labels, coded or represented blind to condition, and analyzed separately from the task and behavioral variables used to validate B. The illustrative content labels above are predicted tendencies, not categories supplied to participants or criteria used to define the bottleneck.

A semantic-yoked comparison can make the strongest reconstruction alternative concrete. For example, a yoked trial can display the same three route cards and choice-oriented language as a selection-limited trial while making one route clearly superior, so independent criteria indicate that selection is easy; implementation can then be made limiting by withholding the same trial-specific control mapping used in the specification condition. If selection-like reports appear whenever the choice semantics are visible despite the absence of independently validated selection dominance, the task-state reconstruction route remains sufficient. If report organization instead follows the independently validated limiting relation across such yoked trials, after generic demand is accounted for, the result is more specifically consistent with the You Are Here correspondence. Neither pattern deductively proves or disproves pre-prompt phenomenality; the purpose is to make the competing routes empirically nonequivalent.

The worked paradigm therefore has explicit loss conditions. It supports only bottleneck-relative reportable content if R_F discriminates independently validated B beyond generic demand. The stronger phenomenal interpretation additionally requires that explicit task-state diagnosis and semantic-yoked conditions not fully reproduce the referential pattern, with the strongest test remaining distributional relocation when B changes within a continuing task. The strong version is weakened or falsified if the intended B classes fail independent validation, if the semantic/metacognitive alternatives absorb the pattern, or if a clean change in B produces no corresponding relocation of focal-content indicators.

9.3 Measuring phenomenal content

The dependent construct must distinguish functional content from generic demand without defining phenomenal categories directly from the manipulated bottleneck labels. At the same time, no verbal probe provides direct access to pre-prompt phenomenal focality. The measurement strategy must therefore distinguish the theoretical variable Φ_F from its observable report-based indicator.

A primary measurement remains sparse, immediate, open-ended experience sampling using a neutral prompt such as:

“What were you trying to figure out immediately before this prompt?”

The purpose of the prompt is not to ask participants where the cognitive system is failing. It is to sample the object or relation around which the participant reports that the immediate task problem was organized.

Reports should be pooled without experimental condition labels. Their content should then be represented or coded by raters blind to condition according to the expressed object or relation of concern. Ideally, an initial exploratory subset would be used to derive a coding scheme, after which the codebook would be frozen before preregistered confirmatory analysis on held-out observations.

Let R_F denote this blinded representation of immediate report content. R_F is an observable indicator relevant to Φ_F ; it is not assumed to be identical to the pre-prompt phenomenal field.

Generic reports such as:

- “this is difficult,”
- “I am confused,”
- “I am stuck,”
- “I am unsure,”

should be represented separately as generic-demand content. Such reports may be psychologically meaningful, but they do not identify what the focal problem is functionally about.

By contrast, reports may differ in referential structure. Without presupposing the final coding scheme, examples could include concern with available possibilities, relations among alternatives, implementation details, feedback, discrepancy, or other independently identified content dimensions.

The critical distinction remains:

more difficult

versus:

difficult in a systematically different way with respect to what the problem concerns .

Only the latter bears directly on the You Are Here hypothesis. But referential differentiation in R_F alone establishes bottleneck-relative reportable content more securely than it establishes pre-prompt phenomenal organization. A further identification step is therefore required.

9.3.1 Distinguishing pre-prompt experiential content from task-state reconstruction

The neutral probe reduces theory-laden prompting, but it does not prevent a participant from reconstructing an answer from the explicit local task state after the probe appears. Condition-blind raters cannot remove that participant-level route. The present design therefore treats phenomenal identification as a triangulation problem rather than as a property of a single question.

A minimal convergent strategy would combine three elements.

First, the primary reports should remain temporally proximal, sparse, and unpredictably sampled so that extended retrospective narration is minimized. A second neutral report format may be interleaved on separate trials where useful—for example, asking what, if anything, was most present in mind immediately before the probe—so that a result is not tied to the wording “trying to figure out.” Any such secondary format must remain theory-neutral and must be coded with the same condition blindness.

Second, the study should separately estimate **explicit task-state diagnosis/reportability**. On separate trials or in separate probe blocks, participants can be asked to identify what the task objectively requires next or what information is currently needed for progress, without treating that answer as a phenomenal report. This estimates how strongly the local task structure itself licenses generation-, selection-, specification-, or monitoring-like descriptions.

Third, the design should include semantic-control or yoked trials where practicable. Such trials would preserve the local cues and explicit problem semantics that make a particular report inferable while altering whether independent behavioral/task criteria establish that the corresponding relation is actually the dominant bottleneck. The exact implementation depends on the eventual paradigm; the identification principle is that task-state semantics and independently validated bottleneck dominance must be partially dissociable somewhere in the design.

These controls yield two different evidential targets. **Bottleneck-relative reportable task representation** is established if R_F discriminates bottleneck conditions but the same differentiation is reproduced by explicit task-state diagnosis or semantic-yoked controls. **Evidence more specifically consistent with bottleneck-relative phenomenal organization** requires convergence of temporally proximal report measures with independently validated B , while the predicted referential differentiation is not fully reproduced by explicit task-state diagnosis/reportability alone and is reduced or absent when the same report semantics are available without the same dominant bottleneck.

No combination of reports makes phenomenology directly observable, and the manuscript does not claim otherwise. The purpose of triangulation is to make the phenomenal interpretation more diagnostic by ruling out a stronger nonphenomenal route that can otherwise generate the same condition-specific words.

9.3.2 Competing routes to the same report

Condition-specific reports are not diagnostic by themselves because at least three causal routes can generate them. The experiment earns evidential value only where independent B validation and the control structure force these routes to make different predictions.

Candidate route	How condition-specific R_F can arise	Discriminator in the proposed design
You Are Here	manipulation -> independently validated B -> latent Φ_F -> observed R_F	R_F tracks validated B beyond generic demand, is not fully reproduced when the same report-relevant semantics occur without the same B, and relocates distributionally when B relocates in a clean dynamic test.
Semantic reconstruction	manipulation -> explicit task state / obvious local semantics -> rapid post-probe inference -> R_F	Separate task-state diagnosis or semantic-yoked trials reproduce the report pattern even when independent criteria do not establish the corresponding bottleneck. In that case the strong phenomenal interpretation is not identified.
Process-specific metacognition	manipulation -> fluency, conflict, uncertainty, or other process cues -> metacognitive feeling -> R_F	Process-cue and metacognitive measures account for the report differentiation without residual leverage from independently validated B or its relocation. You Are Here retains residual value only if the dominant-bottleneck relation adds discriminating structure beyond this account and the semantic controls.

These controls do not make Φ_F directly observable and do not convert classifier accuracy into a measure of phenomenality. They instead narrow the observational equivalence by testing whether the same R_F pattern is obtainable through routes that do not require the proposed B- Φ_F correspondence.

9.4 Controlling generic demand

The proposal explicitly predicts a relation beyond generic demand.

Let X include measures such as:

subjective difficulty ,

uncertainty ,

effort ,

and

response latency ,

with additional demand measures included where justified by the final paradigm.

Perfect equality across conditions is unlikely to be achievable.

The relevant requirement is instead that generic demand be measured sufficiently well to test whether referential report differentiation remains after those variables are accounted for.

The theoretical criterion remains:

$$I(B; \Phi_F \mid X, A) > 0.$$

As emphasized in Section 6, this is a compact predictive/dependence criterion rather than a causal estimand. It also does not, by itself, distinguish phenomenal organization from semantic reconstruction of the local task state.

Within a single well-controlled experimental architecture, a practical first-stage test asks whether the blinded immediate-report representation R_F contains information about bottleneck condition beyond that contained in generic-demand variables.

A confirmatory analysis could therefore compare at least two predictive structures.

The first would use generic-demand variables alone:

$$X \rightarrow B.$$

The second would add independently coded immediate report content:

$$X + R_F \rightarrow B.$$

If the second representation discriminates bottleneck condition in held-out observations beyond what is achievable from generic demand alone, the study has established bottleneck-relative **reportable content** beyond generic demand. The stronger inference to Φ_F additionally requires the triangulation described in Section 9.3.1: the report-content effect must not be fully generated by explicit task-state diagnosis or semantic-yoked controls.

The manuscript does not commit to one statistical estimator. Nor does it treat task-state semantics as merely another growing list of covariates in X . The inferential requirement is design based: bottleneck identity must predict report-content structure beyond generic demand, and the remaining report pattern must survive a serious attempt to separate pre-prompt experiential organization from what participants can construct from explicit local problem structure at report time.

9.5 The dynamic test

The strongest experiment would go beyond between-condition comparisons.

A continuing task could be structured so that the dominant limiting relation changes during the same episode.

For example, a participant might initially lack a viable candidate, later possess several candidates but be unable to resolve among them, and later still select one while remaining unable to specify its implementation.

The predicted trajectory is:

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j.$$

Immediate probes placed close to these transitions would test whether the distribution or referential organization of report content changes with the limiting relation rather than merely reflecting stable task identity.

This dynamic version is especially diagnostic because many alternative explanations can account for different reports across different tasks. However, dynamic relocation is not sufficient by itself: the explicit local task state can change at the same transition and can therefore drive report reconstruction. The strongest version of the design should include transitions or yoked comparisons in which explicit task-state diagnosis and independently validated bottleneck dominance are at least partially dissociated.

Evidence more specifically consistent with You Are Here would then consist of a distributional relocation of task-relevant immediate report content with the dominant bottleneck that is not fully reproduced by the corresponding change in explicit task-state reportability.

9.6 Outcomes

The proposed experiment admits several clear outcomes.

A result consistent with bottleneck-relative **reportable task representation** would show that independently coded immediate report content R_F discriminates bottleneck identity after generic demand is accounted for.

Evidence more specifically consistent with You Are Here would additionally survive the semantic-reconstruction controls: the immediate referential pattern would track independently validated bottleneck dominance beyond what is reproduced by explicit task-state diagnosis/reportability, with the strongest support coming from distributional within-task relocation under partial dissociation of bottleneck dominance from task-state semantics.

A generic-demand result would show that conditions differ in difficulty, effort, uncertainty, or response latency, but that referential report structure contributes no additional discrimination.

A task-state reconstruction result would show condition-specific immediate reports, but equivalent or near-equivalent differentiation would be generated by explicit task-state diagnosis or semantic-yoked controls. That outcome would support bottleneck-relative reportable problem representation without identifying bottleneck-relative phenomenal organization.

A retrospective-reconstruction result would show differentiated explanations retrospectively but little or no corresponding differentiation in immediate reports.

An architectural failure would occur if the manipulated bottlenecks cannot be distinguished independently enough to define B .

A dynamic failure would occur if the dominant limiting relation changes while the distribution or referential organization of the focal phenomenal-content indicators remains functionally unchanged in a clean design.

The last result is particularly important. It would contradict the strongest form of the model rather than merely weaken its effect size.

The proposed study is therefore not intended to demonstrate that unresolved processing becomes conscious.

That broader proposition is unnecessary to the hypothesis and substantially represented in prior literature.

The purpose is narrower:

Determine whether moving the functional location of unresolved progression moves what the immediate conscious problem is about.

10. Scope and Identification Limits

The You Are Here hypothesis concerns a restricted correspondence between a functionally limiting relation and task-relevant focal phenomenal content.

Several boundaries follow from that formulation.

These boundaries constrain what evidence can establish the model and what conclusions could legitimately be drawn if it were supported.

10.1 Limited introspective access

The hypothesis does not require people to identify the cognitive architecture generating their experience.

A participant may report uncertainty about alternatives without knowing that unresolved selection is the relevant functional limitation.

Conversely, a participant may explicitly diagnose the problem incorrectly.

For this reason:

$$B \neq M,$$

where M denotes explicit metacognitive diagnosis.

Phenomenal reports should therefore be interpreted as evidence about experienced content, not as authoritative descriptions of underlying computational organization.

10.2 Reports do not provide direct access to experience

The proposed empirical strategy relies on behavioral reports as observable indicators of task-relevant phenomenal content.

Such reports are imperfect.

They may omit content, compress it linguistically, alter ongoing processing, or incorporate reconstruction between experience and response.

Immediate and sparse sampling can reduce some of these problems but cannot eliminate them.

The model therefore does not claim direct measurement of phenomenal states independent of report. In the empirical strategy, R_F denotes coded immediate report content and is treated as an indicator relevant to, but not identical with, the theoretical Φ_F .

The neutral prompt and condition-blind coding cannot by themselves establish pre-prompt phenomenal focality, because a participant may construct a report from explicit local task-state knowledge after the probe appears. The proposed semantic-diagnosis and yoked-control strategy is therefore part of the identification argument, not optional methodological decoration.

The model predicts a systematic correspondence in observable reportable content under conditions designed to minimize retrospective reconstruction and, more specifically, to test whether task-state/report construction can fully generate the observed referential pattern.

Failure to obtain such a correspondence would remain evidence against the empirically formulated model even if unreported phenomenal differences were logically possible. Likewise, if the observable correspondence is fully reproduced by task-state reconstruction controls, the stronger phenomenal interpretation is not identified. Otherwise the proposal would cease to be falsifiable.

10.3 Retrospective reconstruction

Temporal proximity is central because retrospective explanation can mimic the predicted mapping.

Participants may infer after the fact that they “must have been trying to decide” or “must have been trying to figure out how” based on knowledge of their own behavior.

Such narratives would not establish that the corresponding relation organized focal experience during the unresolved episode.

The strongest claim therefore concerns content sampled during or immediately adjacent to unresolved processing.

If only delayed reports display the predicted differentiation, the evidence supports retrospective reconstruction more readily than online bottleneck-relative phenomenal organization.

10.4 Functional relations need not form a serial chain

Candidate generation, selection, specification, monitoring, and related processes may interact recurrently and hierarchically.

You Are Here does not posit:

$$B_1 \rightarrow B_2 \rightarrow B_3$$

as a compulsory sequence.

A selected option may prove impossible to specify, reopening generation.

Monitoring may reveal new alternatives.

Selection may alter the information required for specification.

Higher-order goals may constrain lower-level candidate availability before any explicit choice occurs.

Accordingly, the term *location* refers to functional location within an organized action system, not necessarily temporal position in a serial pipeline.

The hypothesis is testable wherever functionally distinguishable limiting relations can be identified, regardless of whether the architecture as a whole is serial, recurrent, hierarchical, or partially parallel.

10.5 Multiple simultaneous bottlenecks

Real tasks may contain more than one active constraint.

The present minimal model is easiest to test when one relation is sufficiently dominant to serve as the experimentally identifiable bottleneck.

It does not assert that every moment of action contains exactly one uniquely limiting transition.

Where multiple relations constrain progression simultaneously, the variable B may become ambiguous or multidimensional.

Such cases may require a richer representation than the present formulation.

They cannot be used to rescue a failed clean test.

The initial empirical claim should stand or fall in conditions where bottleneck identity is sufficiently identifiable.

10.6 Attention and consciousness

A bottleneck may alter attention without necessarily altering phenomenal content, and changes in phenomenal content may be accompanied by changes in attention.

The present model does not provide an independent theory of their relationship.

An empirical correspondence between B and reported Φ_F would therefore not show that the bottleneck causes consciousness, nor would it establish whether attentional selection mediates the effect.

The relevant claim is restricted to the organization of task-relevant conscious content as operationalized through immediate report.

10.7 No inference to neural mechanism

The hypothesis is formulated at a functional and phenomenal level.

Even strong evidence for:

$$B_i \rightarrow \Phi_i$$

would not identify the neural mechanism that realizes either variable.

Workspace dynamics, recurrent processing, hierarchical predictive control, metacognitive monitoring, attentional selection, or other mechanisms might contribute to an observed correspondence.

The proposed experiment is not designed to adjudicate among these possibilities.

You Are Here should therefore not be presented as a neural theory of consciousness.

10.8 Common-architecture approximation

The conditioning term A expresses an important idealization.

No experimental manipulation leaves every aspect of an action system unchanged.

Moving a bottleneck will necessarily alter some downstream and possibly upstream processing.

The requirement is therefore not literal identity of the entire system.

It is sufficient similarity to rule out the trivial explanation that different phenomenal reports arise because participants are performing categorically different tasks.

The quality of evidence will depend partly on how convincingly a design preserves the relevant architecture while changing the dominant limiting relation.

10.9 Generic demand is not exhausted by a short list

Difficulty, uncertainty, effort, conflict, and response latency are useful demand variables, but they should not be treated as an exhaustive ontology of generic demand.

Depending on the paradigm, familiarity, surprise, time pressure, option-set size, error likelihood, fluency, evidence strength, or other variables may affect both behavior and phenomenal report.

The theoretical commitment is therefore not to a fixed list of covariates.

It is to the distinction between:

generic demand

and:

referential content identity .

A convincing experiment must measure plausible demand alternatives well enough that residual content differentiation is interpretable.

10.10 Content categories are measurement constructs

Phenomenal-content categories such as “possibilities,” “alternatives,” “implementation,” or “feedback” should not be treated as natural kinds merely because they are theoretically intuitive.

The operationalization deliberately requires content structure to be derived independently of condition labels and then tested on held-out observations.

If independently derived report structure fails to resemble the illustrative mappings proposed in Section 7, the data rather than the illustration should govern interpretation.

The examples explain the model.

They do not define successful evidence in advance.

10.11 Scope of the phenomenal claim

You Are Here concerns the **task-relevant focal phenomenal field**.

It does not propose that all concurrent conscious content is determined by a bottleneck.

A person may simultaneously experience bodily sensations, background emotion, environmental features, autobiographical thoughts, or unrelated concerns.

The hypothesis therefore concerns preferential organization of the focal task problem, not total phenomenal consciousness.

10.12 Identification rather than metaphysical explanation

Finally, even a successful test would establish less than a complete theory of consciousness.

The proposed correspondence would show that functional bottleneck location contains predictive information about the organization of focal task experience under specified conditions.

It would not establish why phenomenal consciousness exists, why particular states are conscious rather than unconscious, or why conscious experience has qualitative character at all.

Those questions lie outside the scope of the model.

The intended contribution is narrower:

identification and formalization of a testable cross-level correspondence

between the functional organization of ongoing action and the organization of its task-relevant phenomenal problem.

11. Implications for Subjectivity and Agency

If supported, the correspondence would bear directly on human subjectivity in goal-directed action and on agency, where complex action production contrasts with comparatively limited focal conscious content. Goals, alternatives, specifications, predicted outcomes, feedback, and control adjustments may all contribute to behavior while only a small portion becomes phenomenally focal; You Are Here proposes one possible organizing relation within that mismatch.

Rather than treating task-relevant conscious experience as a transparent representation of the entire action architecture, the model predicts that focal experience may preferentially expose the relation at which progression is currently unresolved.

Under this interpretation, the larger system may contain richer causal organization than the immediate phenomenal problem, making the focal field systematically selective rather than a complete map of action production.

This possibility has several implications for agency.

First, salient content need not identify all processes contributing to behavior. If generation has succeeded and selection is limiting, alternatives may dominate focal experience while their generative processes remain peripheral; if specification is limiting, the selected goal may remain stable while implementation becomes focal.

Second, conscious uncertainty may be local rather than global: a person can know the goal while not knowing which action to select, or know the intended action while not knowing how to implement it. Treating these as undifferentiated uncertainty obscures their possible functional organization.

Third, explicit reports of agency should not automatically be interpreted as transparent causal explanations.

The distinction among B , Φ_F , and M permits phenomenal organization to correspond systematically to an unresolved functional relation even when explicit metacognitive diagnosis remains incomplete or incorrect.

If supported, the model would therefore encourage a more local interpretation of some conscious action phenomena.

What is phenomenally focal may reveal something about where progression is unresolved without revealing the entire causal history of the action.

This remains a conditional implication.

The present manuscript does not establish that this is how agency is organized.

It identifies an empirical correspondence whose confirmation would make that interpretation more plausible.

11.1 General-Psychology Significance

The broader general-psychology significance lies in relating two descriptions of the same goal-directed process: its functional organization and the organization of lived task content.

At one descriptive level is the functional organization of action:

$$A,$$

within which different relations can become limiting.

At another is the organization of the task-relevant phenomenal field:

$$\Phi_F.$$

The proposed correspondence asks whether a local constraint defined at the first level predicts organization observed at the second.

This is not a reduction of phenomenology to functional architecture.

Nor is it a claim that functional descriptions and phenomenal descriptions are interchangeable.

It is a proposed cross-level predictive correspondence: the identity of B is hypothesized to carry information about the organization of Φ_F under the stated architectural and demand restrictions. No reciprocal causal influence or equivalence between the two variables is implied.

If supported, such a correspondence would connect psychological accounts of goal-directed action to the organization of local experiential content without reducing phenomenology to functional architecture or treating conscious experience as a transparent representation of the full system state.

The larger causal map could remain distributed across the action architecture while the focal phenomenal field preferentially reflects the locally unresolved relation.

11.2 Possible applications

The framework may eventually be relevant to domains in which people report pronounced variation in the kinds of difficulty encountered during goal-directed action.

Agency is the most direct theoretical application because the proposal concerns what becomes phenomenally focal while action is being generated, selected, specified, and monitored.

Other applications, including individual differences in executive control or conditions such as ADHD, would require separate empirical development.

Such domains cannot serve as evidence for the general correspondence merely because their phenomenology appears compatible with it.

They are therefore best treated as potential later applications rather than as support for the present theory.

The immediate empirical priority is more basic:

Determine whether the proposed bottleneck-content correspondence exists under controlled conditions at all.

12. Conclusion

Goal-directed action contains functionally distinct relations that can limit progression for different reasons. Prior research already distinguishes generation, selection, specification, and monitoring; shows impasse-specific local problem organization; links unresolved conditions to broader control and conscious relevance; and supports structured phenomenal task experience. None of this demonstrates You Are Here.

The unresolved question is narrower.

When a substantially common action architecture is limited at different functional relations, does the task-relevant phenomenal problem differ accordingly?

Formally, the minimal proposal is:

$$I(B; \Phi_F | X, A) > 0.$$

Its strongest dynamic expression is:

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j.$$

This is shorthand for a predicted distributional shift in the referential organization of task-relevant focal phenomenal content when the dominant bottleneck changes, not a deterministic one-to-one transition on every trial.

The prediction concerns identity rather than intensity: shifting from generation to selection or specification should alter what the unresolved problem is about, not merely its difficulty. That prediction remains undemonstrated.

A purpose-built experiment could test it by manipulating the dominant limiting relation within a substantially common task architecture, sampling immediate task-related experience through neutral open-ended reports, deriving report-content structure independently of experimental condition, and triangulating that structure against explicit task-state diagnosis and semantic-reconstruction controls. The first empirical target is bottleneck-relative reportable content; the stronger phenomenal interpretation requires that this pattern not be fully generated by what participants can infer from explicit local task structure at report time.

Several outcomes would reject or sharply reduce the proposal.

If different bottlenecks produce only undifferentiated difficulty, if condition-specific reports are fully explained by task-state/report reconstruction, if phenomenal differentiation appears only retrospectively, if bottleneck location cannot be independently identified, if the strongest metacognitive comparators fully absorb the result, or if the distribution of focal content fails to relocate when the dominant bottleneck moves, then the proposed correspondence does not survive in its strong form.

The hypothesis therefore accepts a narrow burden of proof. For theoretical psychology, its value is not a new mechanism of consciousness but a falsifiable bridge between functional organization and lived task content. The larger action system may contain the whole causal map while focal task experience exposes only a small part; You Are Here proposes that this exposed part may preferentially correspond to the relation currently preventing progress.

References

Baars, B. J. (2002). The conscious access hypothesis: Origins and recent evidence. *Trends in Cognitive Sciences*, 6(1), 47–52. [https://doi.org/10.1016/S1364-6613\(00\)01819-2](https://doi.org/10.1016/S1364-6613(00)01819-2)

- Bor, D., & Seth, A. K. (2012). Consciousness and the prefrontal parietal network: Insights from attention, working memory, and chunking. *Frontiers in Psychology*, 3, 63. <https://doi.org/10.3389/fpsyg.2012.00063>
- Botvinick, M. M. (2008). Hierarchical models of behavior and prefrontal function. *Trends in Cognitive Sciences*, 12(5), 201–208. <https://doi.org/10.1016/j.tics.2008.02.009>
- Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological Review*, 108(3), 624–652. <https://doi.org/10.1037/0033-295X.108.3.624>
- Cisek, P. (2007). Cortical mechanisms of action selection: The affordance competition hypothesis. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362(1485), 1585–1599. <https://doi.org/10.1098/rstb.2007.2054>
- Dehaene, S., & Naccache, L. (2001). Towards a cognitive neuroscience of consciousness: Basic evidence and a workspace framework. *Cognition*, 79(1–2), 1–37. [https://doi.org/10.1016/S0010-0277\(00\)00123-2](https://doi.org/10.1016/S0010-0277(00)00123-2)
- Desender, K., Teuchies, M., Gonzalez-Garcia, C., De Baene, W., Demanet, J., & Brass, M. (2021). Metacognitive awareness of difficulty in action selection: The role of the cingulo-opercular network. *Journal of Cognitive Neuroscience*, 33(12), 2512–2522. https://doi.org/10.1162/jocn_a_01773
- Desender, K., Van Opstal, F., & Van den Bussche, E. (2017). Subjective experience of difficulty depends on multiple cues. *Scientific Reports*, 7, 44222. <https://doi.org/10.1038/srep44222>
- Fedor, A., Szathmáry, E., & Öllinger, M. (2015). Problem solving stages in the five square problem. *Frontiers in Psychology*, 6, 1050. <https://doi.org/10.3389/fpsyg.2015.01050>
- Fernández Velasco, P., & Loev, S. (2025). Metacognitive feelings: A predictive-processing perspective. *Perspectives on Psychological Science*, 20(4), 691–713. <https://doi.org/10.1177/17456916231221976>
- Gurwitsch, A. (1964). *The field of consciousness*. Duquesne University Press.
- Kaiser, S., Simon, J. J., Kalis, A., Schweizer, S., Tobler, P. N., & Mojzisch, A. (2013). The cognitive and neural basis of option generation and subsequent choice. *Cognitive, Affective, & Behavioral Neuroscience*, 13(4), 814–829. <https://doi.org/10.3758/s13415-013-0175-5>
- Kalis, A., Kaiser, S., & Mojzisch, A. (2013). Why we should talk about option generation in decision-making research. *Frontiers in Psychology*, 4, 555. <https://doi.org/10.3389/fpsyg.2013.00555>
- Knoblich, G., Ohlsson, S., Haider, H., & Rhenius, D. (1999). Constraint relaxation and chunk decomposition in insight problem solving. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25(6), 1534–1555. <https://doi.org/10.1037/0278-7393.25.6.1534>
- Laird, J. E. (2012). *The Soar cognitive architecture*. MIT Press. <https://doi.org/10.7551/mitpress/7688.001.0001>
- Lewis, M. D., & Todd, R. M. (2005). Getting emotional: A neural perspective on emotion, intention, and consciousness. *Journal of Consciousness Studies*, 12(8–10), 210–235.

Proust, J. (2024). Affordances from a control viewpoint. *Philosophical Psychology*, 37(7), 1590–1614. <https://doi.org/10.1080/09515089.2023.2274489>

Rietveld, E., Denys, D., & van Westen, M. M. (2018). Ecological-enactive cognition as engaging with a field of relevant affordances: The Skilled Intentionality Framework (SIF). In A. Newen, L. de Bruin, & S. Gallagher (Eds.), *The Oxford handbook of 4E cognition* (pp. 41–70). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780198735410.013.3>

Ross, W., & Arfini, S. (2024). Impasse-driven problem solving: The multidimensional nature of feeling stuck. *Cognition*, 246, 105746. <https://doi.org/10.1016/j.cognition.2024.105746>

Yoshimi, J. (2017). The phenomenology of problem solving. *Grazer Philosophische Studien*, 94(3), 391–409. <https://doi.org/10.1163/18756735-09403006>

Yoshimi, J., & Vinson, D. W. (2015). Extending Gurwitsch's field theory of consciousness. *Consciousness and Cognition*, 34, 104–123. <https://doi.org/10.1016/j.concog.2015.03.017>